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[Home](#) → [Medical Encyclopedia](#) → Anti-insulin antibody test

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Anti-insulin antibody test

The anti-insulin antibody test checks to see if your body has produced antibodies against insulin.

Antibodies are proteins the body produces to protect itself when it detects anything foreign, such as a virus or transplanted organ.

How the Test is Performed

A blood sample is needed.

How to Prepare for the Test

No special preparation is needed.

How the Test will Feel

When the needle is inserted to draw blood, some people feel moderate pain. Others feel only a prick or stinging. Afterward, there may be some throbbing or a slight bruise. This soon goes away.

Why the Test is Performed

This test may be performed if:

- You have or are at risk for type 1 diabetes
- You appear to have an allergic response to insulin
- Insulin no longer seems to control your diabetes
- You are taking insulin to control your diabetes and your blood sugar level varies a lot, with both high and low numbers that can't be explained by the food you are eating relative to the timing of your insulin injections

Normal Results

Normally, there are no antibodies against insulin in your blood. Antibodies can be found in the blood of many people who are taking insulin to control diabetes.

Normal value ranges may vary slightly among different labs. Some labs use different measurements or test different samples. Talk to your health care provider about the meaning of your specific test results.

What Abnormal Results Mean

The presence of IgG and IgM antibodies against insulin can be part of the testing that diagnoses you with type 1 diabetes.

If you develop anti-insulin antibodies, insulin may not work as well, or it may not work at all. As a result, your blood sugar can be very high. Or the antibodies can also put you at risk for low blood sugar.

Your provider may suggest a different form of insulin to which your body may be less likely to form antibodies.

If the test shows a high level of IgE antibody against insulin, your body has developed an allergic response to the insulin.

- This could put you at risk for skin reactions where you inject insulin. You can also develop more severe reactions that affect your blood pressure or breathing.
- Medicines, such as antihistamines or low-dose injectable steroids, may help. Or you may need a treatment process called desensitization or a treatment to remove the antibodies from your blood.

Risks

There is little risk involved with having your blood taken. Veins and arteries vary in size from one person to another and from one side of the body to the other. Obtaining a blood sample from some people may be more difficult than from others.

Other risks of having blood drawn are slight, but may include:

- Excessive bleeding
- Fainting or feeling lightheaded
- Multiple punctures to locate veins
- Hematoma (blood building up under the skin)
- Infection (a slight risk any time the skin is broken)

Alternative Names

Insulin antibodies - serum; Insulin Ab test; Insulin resistance - insulin antibodies; Diabetes - insulin antibodies

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